



Verbal and numeric probabilities differentially shape decisions

Robert N. Collins, David R. Mandel & Brooke A. MacLeod

To cite this article: Robert N. Collins, David R. Mandel & Brooke A. MacLeod (2024) Verbal and numeric probabilities differentially shape decisions, *Thinking & Reasoning*, 30:1, 235-257, DOI: 10.1080/13546783.2023.2220971

To link to this article: <https://doi.org/10.1080/13546783.2023.2220971>



© 2023 Crown Copyright. Published by Informa UK Limited, trading as Taylor & Francis Group.



Published online: 14 Jun 2023.



Submit your article to this journal [↗](#)



Article views: 1099



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 1 View citing articles [↗](#)

BRIEF REPORT

 OPEN ACCESS



Verbal and numeric probabilities differentially shape decisions

Robert N. Collins , David R. Mandel  and Brooke A. MacLeod

Intelligence, Influence and Collaboration Section, Toronto Research Centre, Defence Research and Development Canada, Toronto, Ontario, Canada

ABSTRACT

Experts often communicate probabilities verbally (e.g., *unlikely*) rather than numerically (e.g., 25% chance). Although criticism has focused on the vagueness of verbal probabilities, less attention has been given to the potential unintended, biasing effects of verbal probabilities in communicating probabilities to decision-makers. In four experiments ($Ns=201, 439, 435, 696$), we showed that probability format (i.e., verbal vs. numeric) influenced participants' inferences and decisions following a hypothetical financial expert's forecast. We observed a format effect for low probability forecasts: verbal probabilities were interpreted more pessimistically than numeric equivalents. We attributed the difference to *directionality*, a linguistic property that biases attention toward an outcome. In the high-probability conditions, the directionality of verbal and numeric probabilities aligned (both were positive), whereas they differed in the low-probability conditions (verbal probabilities were more negative). Participants inferred recommendations congruent with the communicated direction and these inferences mediated the effect of probability format on decisions.

ARTICLE HISTORY Received July 15, 2022; Accepted May 20, 2023

KEYWORDS Verbal probability; numeric probability; directionality; recommendation; decision making

Introduction

Expert judgment is frequently relied upon to inform consequential decisions. A physician's prognosis affects a patient's choice of treatment programs (Lipkus, 2007; Mazur & Merz, 1994). A financial advisor's market forecast affects investors' strategies (Juanchich et al., 2012). A forensic analyst's communication of uncertainty about evidence will shape the conclusions of judges and juries (Ligertwood & Edmond, 2012). Furthermore,

CONTACT Robert N. Collins  robert.collins3@ecn.forces.gc.ca  Intelligence, Influence and Collaboration Section, Toronto Research Centre, Defence Research and Development Canada, Toronto, Ontario, Canada. This article has been corrected with minor changes. These changes do not impact the academic content of the article. © 2023 Crown Copyright. Published by Informa UK Limited, trading as Taylor & Francis Group. This is an Open Access article distributed under the terms of the Creative Commons Attribution License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

expert forecasts inform all domains of public policymaking (Funtowicz & Ravetz, 1990), from intelligence analysis to the IPCC's climate projections (Ho et al., 2015). In these and other areas, effective decision-making depends on the clear and unbiased communication of probabilities.

A great deal of research, however, suggests that contemporary methods of communicating probabilities run counter to these aims (for a review, see Dhami & Mandel, 2022). Organizations and individual experts often prefer to express probability with verbal terms such as *likely* instead of numeric point estimates such as 70% (Dhami & Mandel, 2021; Erev & Cohen, 1990; Juanchich & Sirota, 2020; Mandel & Irwin, 2021c). The purported benefits are that verbal probabilities avoid over-precision and are easier to produce than strictly quantitative judgments (Wallsten et al., 1993). However, compared to numeric probabilities, verbal probabilities are ineffective at communicating uncertainty levels clearly (Dhami & Mandel, 2022). There is large variability in their interpretation between individuals (Barnes, 2016; Beyth-Marom, 1982; Budescu et al., 2009) and even among individuals across differing contexts (Brun & Teigen, 1988; Mellers et al., 2017; Wallsten & Budescu, 1995). The vagueness of verbal probabilities can result in poorer comprehension, even when practitioners and organizations try to clarify verbal terms with stipulated numerical ranges (Budescu et al., 2014; Ho et al., 2015; Mandel & Irwin, 2021a, 2021b; Wintle et al., 2019). Receivers of probabilistic assessments tend to rate numeric probabilities as conveying clearer information about probability levels than verbal probabilities (Collins & Mandel, 2019).

In addition to vagueness, the linguistic properties of verbal quantifiers may “leak” information that influences inferences (Honda & Yamagishi, 2017; McKenzie & Nelson, 2003). For instance, verbal probabilities may indirectly communicate information about reference points that influence behavior. Sher and McKenzie (2006) asked participants to equalize the water level between two cups, one full and one empty, and to hand the experimenter one of the cups afterward. The cup handed to the experimenter depended on whether the experimenters requested the ‘half-empty’ or ‘half-full’ cup: 69% percent of participants in the former condition supplied the previously full cup, whereas only 46% of participants in the latter condition supplied the previously full cup. Despite the apparent equivalence of these requests, the phrasing appears to have leaked information about the experimenter's expectations and influenced their behavior.¹

In studies of verbal probabilities, information leakage occurs *via* the *directionality* of probability expressions (Brun & Teigen, 1988; Teigen & Brun, 1995, 1999).² For example, if a friend said it was *doubtful* they would attend an event you may be less inclined to prepare for their attendance than if they said there was a *small chance*, even though both communicate

¹In most framing research, extensional equivalence is assumed but is not empirically tested (Fisher & Mandel, 2021; Mandel, 2022); hence, our cautious use of the term “apparent”.

²Directionality is sometimes referred to as *polarity* (e.g., Hilton, 2011; Moxey & Sanford, 1993, 2000).

similar probability levels (Honda & Yamagishi, 2006, 2009, 2017; Schmeltzer & Hilton, 2014). *Doubtful* expresses negative directionality and focuses attention on the large chance of non-attendance. Conversely, *a small chance* expresses positive directionality and focuses attention on the possibility of attendance. In either case, the attentional focus reveals the communicator's expectations and may influence the receiver's behavior. Importantly, directionality signals the communicator's expectations *independent* of its probability. This is potentially advantageous when receivers seek behavioral guidance and advice. However, it is problematic if: (1) no recommendation was intended, (2) unbiased, neutral communication is mandated, or (3) the expert's choice of the verbal term—and therefore directionality—is constrained by an arbitrary standard. Under any of these conditions, the integrity of experts' verbal probability forecasts may be compromised.

In practice, *congruent* verbal terms (i.e., positive-directional high-probability terms and negative-directional low-probability terms) are more common and intuitive than *incongruent* verbal terms (i.e., negative-directional high-probability terms and positive-directional low-probability terms; Budescu et al., 2003; Juanchich et al., 2013). For example, terms like *impossible*, *improbable*, *probable*, and *certain* all communicate directionality that matches their position above (positive) or below (negative) an *even chance*, i.e., $p = .50$. By comparison, incongruent terms typically require the use of negation or modifying adjectives—*not impossible*, *not improbable*, *a small possibility*, and *not entirely certain*—phrasing that is often clunky, unnatural, and uncommon in communication. In contrast to verbal terms, numerical probabilities typically communicate positive directionality regardless of the probability level communicated (Teigen & Brun, 2000). Consequentially, the choice of probability format may have a greater effect at the low end of the probability scale: low-probability verbal terms are more likely to be negative (Budescu et al., 2003; Juanchich et al., 2013) and conflict with positive numeric probabilities. This may cause conflicting interpretations between ostensibly equal probabilities, especially for low-probability events where negative directionality verbal terms are common.

Collins and Mandel (2019; see also Jenkins et al., 2017; Jenkins & Harris, 2021) found indirect support for this hypothesis. Participants received an expert investor's forecast that a stock would increase in value. The probability format (verbal vs. numeric) and level (low vs. high) were manipulated between conditions. For low and high verbal probabilities, the researchers used *unlikely* (negatively directional) and *likely* (positively directional), respectively. The researchers used numeric probabilities approximating the terms' average interpretation in the numeric conditions. Participants presented with high-probability forecasts judged the forecaster's credibility similarly regardless of whether they received verbal or numeric forecasts. However, credibility differed by format in the low probability condition. When low-probability forecasts were accurate—namely,

the low-probability event did not happen—the negative verbal forecasts bolstered credibility more than the positive numeric forecasts. Conversely, when low-probability forecasts were inaccurate—namely, the low-probability event happened—the negative verbal forecasts undermined credibility more than numeric forecasts. These results suggest low verbal and numeric probabilities conveyed differing implicit recommendations, which receivers repaid with praise when these recommendations proved to be beneficial and with blame when they proved to be costly. Also problematic was the fact that participants rated verbal probabilities as providing clearer recommendations, even though none was provided.

The present research evaluates the directionality hypothesis directly by examining how probability format and probability level interact to influence the recommendation receiver infer from the expert's forecast, and whether these inferences affected their decisions. Specifically, we hypothesized that the probability format would have a negligible effect on these measures when probability levels were high because both formats typically convey positive directionality. Conversely, we hypothesized that format would have a large effect when probability levels were low because the two formats convey different directionality. Specifically, the verbal probabilities convey negative directionality, but the numeric probabilities convey positive directionality. Therefore, following a low-probability forecast, we expected greater divergence in inferred recommendations and decisions consistent with the proposed differences in directionality. Moreover, we hypothesized that following a low-probability forecast; the effect of probability format on participants' decisions would be mediated by their inferred recommendations. That is, we expected that under low-probability conditions, the probability format selected by an advisor would influence inferred recommendations from that advisor, and participants' decisions would be consistent with the inferred advice.

Experiment 1

Materials, methods, experimental data, and analysis code for all experiments can be found in the companion repository on the Open Science Framework website (Collins et al., 2022; <https://osf.io/evwm8/>).

Materials and methods

Participants

For the first experiment, we aimed to achieve 90% power to detect the effect sizes based on the interactions found in Collins and Mandel (2019), $\eta_p^2 \approx .05$. A G*Power (Faul et al., 2007) sensitivity analysis suggested a sample size requirement of two-hundred and two. Two-hundred and one English-speaking participants (96 male) between 18-60 years of age ($M = 40.7$, $SD = 10.8$) who were U.S. ($n = 97$) or Canadian ($n = 104$) citizens

were recruited from Qualtrics Panels and received compensation from the Panels provider. Our primary concern in sample selection in all experiments was to recruit participants who were fluent English speakers.

Design

We used a 2 (Probability Format: numeric point, verbal) $\times$ 2 (Probability Level: low, high) between-subjects factorial design for the experiment and analyses. As in Collins and Mandel (2019), we used the terms *unlikely* and *likely* for the low- and high-probability verbal forecasts, respectively; and the numeric phrases *about a 25% chance* and *about a 75% chance* for the low- and high-probability numeric point forecasts, respectively. These values are complementary, proximal equivalents of their verbal counterparts (Barnes, 2016; Mosteller & Youtz, 1990; Theil, 2002).

Materials and procedure

The survey was administered *via* the Qualtrics survey platform. After providing consent, participants read the following fictional stock market vignette. The probability format and level manipulations are shown in brackets:

Imagine that you had some money that you were looking to invest in the stock market in the hopes of returning substantial profits. You're looking for stocks that offer favourable odds of substantial profit and you're looking to avoid stocks that don't fit this description. You consult a financial advisor recommended to you by a close friend regarding your investment plan for the next year, and you explain your objectives. Among other information, the advisor draws your attention to a key forecast: **Bayosia Corp's Stock [is unlikely/is likely/has about a 25% chance/has about a 75% chance] to increase over the next year.** [emphasis in original]

Following the vignette, participants answered the following questions in the order shown: (1) "What course of action do you think the advisor is recommending?" (-5 = Definitely do not invest, $+5$ = Definitely do invest), (2) "Given the advisor's forecast, would you choose to invest in Bayosia Corp's stock?" (No, Yes), and (3) "How confident are you in this decision?" (0 = Not at all confident, 10 = Completely confident). Participants were then debriefed.

Results

Inferred recommendation

We conducted a two-way (Probability Format $\times$ Probability Level) analysis of variance (ANOVA) of inferred recommendation. The ANOVA revealed main effects of both probability format, $F(1, 197) = 13.42$, $p < .001$, $\eta_p^2 = .064$, and level, $F(1, 197) = 89.19$, $p < .001$, $\eta_p^2 = .312$, as well as the

predicted interaction effect, $F(1, 197) = 14.41, p < .001, \eta_p^2 = .064$. Supporting the directionality hypothesis, the top row of Figure 1 shows the effect of format was only significant in the low-probability condition, $t[99]=4.58, p < .001, d=0.92$. Participants rated verbal forecasts as a much more pessimistic recommendation against investing ($M=-1.42, SD=2.57$) than comparable numeric point forecasts ($M=0.93, SD=2.56$).

Confidence-weighted decision

We computed a confidence-weighted decision measure (Mandel, 2014; Peters & Levin, 2008) by coding participant investment decisions (no=-1, yes = +1) and multiplying the result by their confidence rating. The derived scale had values ranging from -10 (do not invest, completely confident) to 0 (unsure, not confident) to 10 (invest, completely confident). A two-way (Probability Format×Probability Level) ANOVA on this measure revealed main effects of probability format, $F(1, 197) = 10.75, p = .001, \eta_p^2 = .052$, and level, $F(1, 197) = 65.35, p < .001, \eta_p^2 = .249$, and the predicted interaction effect, $F(1, 197) = 9.85, p = .002, \eta_p^2 = .048$. As the top row of Figure 2 shows, the effect of format was, once again, significant only in the

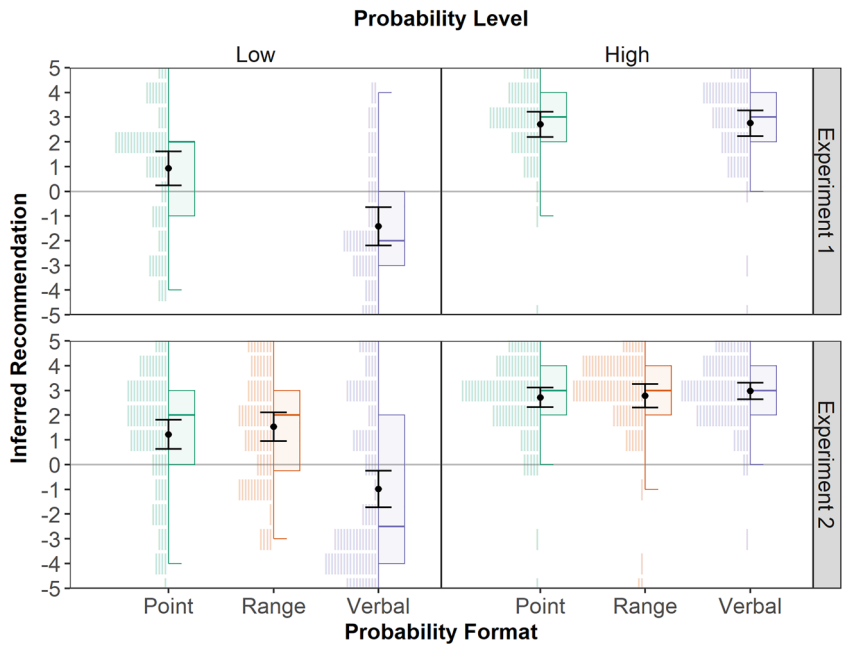


Figure 1. Effect of probability format and probability level on inferred recommendation.

Note: Histograms are sample data and each stroke represents one participant; Box-and-whiskers plots are sample data; error plots are estimated marginal means and 95% CI. The inferred recommendation scale values run from -5 = *Definitely do not invest* to +5 = *Definitely do invest*.

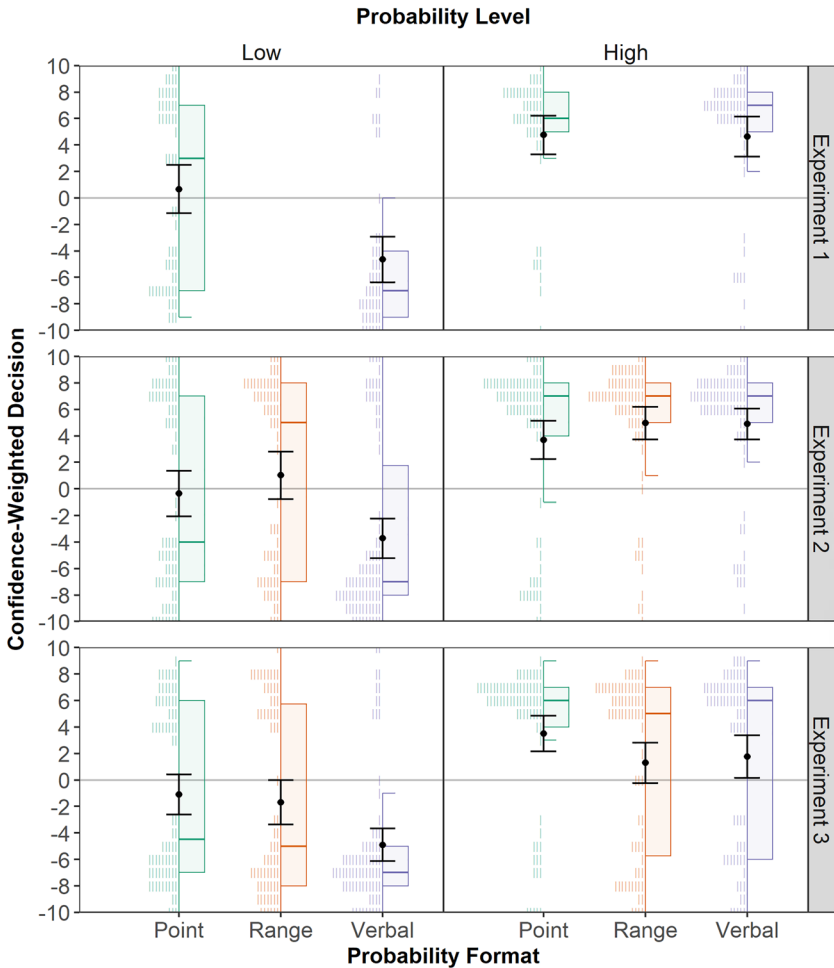


Figure 2. Effect of probability format and probability level on confidence-weighted decisions.

Note: Histograms are sample data and each stroke represents one participant; Box-and-whiskers plots are sample data; error plots are estimated marginal means and 95% CI. The confidence-weighted decision scale is a product of the decision response (no = -1, yes = +1) and the confidence response (0 = Not at all confident, 10 = Completely confident).

low-probability condition, $t(98.73) = 4.24$, $p < .001$, $d = 0.83$. Participants in the verbal condition were much more pessimistic ($M = -4.64$, $SD = 5.76$) than participants in the numeric point estimate condition ($M = 0.66$, $SD = 6.82$).

Mediator analysis

We conducted a mediation analysis of the low-probability forecasts using the PROCESS Procedure (Hayes, 2022) that treated probability format as the predictor variable X , confidence-weighted decision as the outcome

variable *Y*, and inferred recommendation as the mediating variable *M*. The results of the mediation analysis are presented in Table 1. The results show that probability format predicted inferred recommendation (*a*) and that inferred recommendation predicted confidence-weighted decisions (*b*). The total, unmediated effect of probability format on confidence-weighted decisions (*c*) was significant, suggesting a predictive relationship. However, the direct effect of probability format on confidence-weighted decisions was not significantly different from 0 (*c'*). Additionally, the indirect effects analysis (*ab*) shows a correspondingly large decrease in coefficients associated with probability format when the effects of inferred recommendation are partialled out. Combined, the analyses show clear evidence that inferred recommendation largely mediated the effects of probability format on confidence-weighted decisions.

Experiment 2

The results of Experiment 1 supported the directionality account. Crucially, verbal probabilities differ from numeric point estimates in another way: numeric point estimates are precise, verbal probabilities are imprecise. Imprecise estimates may convey greater uncertainty (Dieckmann et al., 2010; Teigen & Brun, 2000; Theil, 2002), and individuals tend to overvalue certainty in decision-making (Tversky & Kahneman, 1986). Participants may simply have been more optimistic following a certainty-instilling numeric point estimate than an uncertainty-instilling verbal probability, particularly when probabilities are low as opposed to high. To that end, we repeated Experiment 1, adding a numeric-range format condition. Like verbal probabilities, numeric ranges are imprecise. However, they have positive directionality like numeric point estimates. This distinction provides a basis to differentiate the effects of directionality and certainty from those of directionality.

Materials and methods

Participants

For Experiment 2, we sought to increase the sensitivity to 99% based on the effect size of the weakest interaction in Experiment 1 ($\eta_p^2 = 0.048$ for

Table 1. Results of mediation analysis of probability format (*X*) on confidence-weighted decisions (*Y*) by inferred recommendation (*M*) for Experiment 1.

Path	Effect	<i>b</i> (95% CI)	SE	β	<i>t</i>	df	<i>p</i>
<i>a</i>	<i>X</i> → <i>M</i>	2.35 (1.33, 3.37)	0.51	.84	4.58	100	< .001
<i>b</i>	<i>M</i> → <i>Y</i>	1.57 (1.18, 1.96)	0.19	.64	8.06	99	< .001
<i>c</i>	<i>X</i> → <i>Y</i>	5.31 (2.77, 7.84)	1.28	.77	4.16	100	< .001
<i>c'</i>	<i>X</i> → <i>Y</i>	1.62 (-0.56, 3.79)	1.09	.24	1.48	99	.143
<i>ab</i>	<i>X</i> → <i>M</i> → <i>Y</i>	3.69 (2.02, 5.56)	0.91	.54			

Note: The indirect effect estimate *ab* is derived from a bootstrapped simulation of 10000 samples.

confidence-weighted decision). A G*Power (Faul et al., 2007) sensitivity power analysis suggested a sample size of four-hundred and twenty-eight participants. Four-hundred and thirty-nine English-speaking participants (219 male) holding Canadian ($n=223$) or U.S. ($n=216$) citizenship between the ages of 18 and 60 ($M=41.6$, $SD=11.6$) were recruited from Qualtrics Panels and received a credit from the Panels provider.

Design, materials, and procedure

The design of Experiment 2 was identical to Experiment 1 except for the addition of the numeric range format, resulting in a 3 (Probability Format: numeric point, numeric range, verbal) $\times$ 2 (Probability Level: low, high) between-subjects design. For materials, we altered the numeric point estimate values from 25% to 30% and from 75% to 70% in the low- and high-probability conditions, respectively. These values more closely align with the membership functions of *unlikely* and *likely* reported by Ho et al. (2015). We created the respective numeric range estimates by adding or subtracting 10 percentage points from the numeric point values, resulting in estimates of a 20-40% chance and a 60-80% chance for the low and high-probability conditions, respectively.

Results

Inferred recommendation

A two-way (Probability Format $\times$ Probability Level) ANOVA on inferred recommendation revealed main effects of both probability format, $F(2, 433) = 10.64$, $p < .001$, $\eta_p^2 = .047$, and level, $F(1, 433) = 100.11$, $p < .001$, $\eta_p^2 = .188$, as well as the predicted interaction effect, $F(2, 433) = 15.34$, $p < .001$, $\eta_p^2 = .066$. As the bottom row of Figure 1 shows, supporting the directionality account, the format effect was only significant in the low-probability condition, $F(2, 216) = 18.17$, $p < .001$, $\eta_p^2 = .144$. Post-hoc Tukey Honest Significant Difference (HSD) revealed that verbal forecasts were inferred to be more pessimistic ($M = -0.99$, $SD = 3.36$) than either numeric point ($M = 1.22$, $SD = 2.52$, $p < .001$) or range ($M = 1.53$, $SD = 2.32$, $p < .001$) forecasts but that the two numeric forecasts did not differ from one another ($p = .794$).

Confidence-weighted decision

We submitted the confidence-weighted decision measure to a two-way (Probability Format $\times$ Probability Level) ANOVA, which revealed main effects of both probability format, $F(2, 433) = 5.12$, $p = .006$, $\eta_p^2 = .023$, and level, $F(1, 433) = 82.71$, $p < .001$, $\eta_p^2 = .160$, as well as the predicted interaction effect, $F(2, 433) = 6.60$, $p = .002$, $\eta_p^2 = .030$. As the middle row of Figure 2

Table 2. Results of mediation analysis of probability format (X) on confidence-weighted decisions (Y) by inferred recommendation (M) for Experiment 2.

Path	Effect	<i>b</i> (95% CI)	SE	β	<i>t</i>	df	<i>p</i>
<i>a</i>	X → M	1.13 (0.67, 1.58)	0.23	.31	4.87	218	< .001
<i>b</i>	M → Y	1.60 (1.35, 1.86)	0.13	.66	12.29	217	< .001
<i>c</i>	X → Y	1.74 (0.60, 2.88)	0.58	.20	3.01	218	.003
<i>c'</i>	X → Y	−0.07 (−0.99, 0.86)	0.47	−0.01	−0.14	217	.886
<i>ab</i>	X → M → Y	1.81 (1.04, 2.62)	0.41	.21			

Note: The indirect effect estimate *ab* is derived from a bootstrapped simulation of 10000 samples.

shows, once again, the effect of format was only significant in the low-probability condition, $F(2, 216) = 8.88, p < .001, \eta_p^2 = .076$. Post-hoc Tukey HSD revealed that confidence-weighted decisions were significantly more pessimistic in the verbal condition ($M = -3.72, SD = 6.77$) than either the numeric point condition ($M = -0.34, SD = 7.35; p = .009$) or the numeric range condition ($M = 1.02, SD = 7.15; p < .001$), but the latter two conditions did not differ ($p = .502$).

Mediator analysis

As in Experiment 1, we conducted a mediation analysis of responses in the low-probability condition that treated probability format as the predictor variable *X*, confidence-weighted decision as the outcome variable *Y*, and inferred recommendation as the mediating variable *M*. The results of the mediation analysis are presented in Table 2. The results again show that probability format predicted inferred recommendation (*a*) and that inferred recommendation predicted confidence-weighted decisions (*b*). The total, unmediated effect of probability format on confidence-weighted decisions (*c*) was significant. However, the direct effect of probability format on confidence-weighted decisions was almost exactly 0, and indeed not significantly different from 0 either (*c'*). Furthermore, the indirect effects analysis (*ab*) again showed a large decrease in coefficients associated with probability format when the effects of inferred recommendation are partialled out. This result replicates Experiment 1 and provides further support for the hypothesis that inferred recommendation largely mediated the relationship between probability format and confidence-weighted decisions.

Experiment 3

Experiments 1 and 2 provided support for the directionality account regardless of the expressed precision of numeric forecasts. However, it is unclear whether the observed format by probability-level interaction on decision-making depended on the prior, explicit instruction to reflect on the inferred recommendation. Experiment 3 addressed this issue by

omitting the recommendation query. We expected a pattern of results comparable to those obtained in Experiments 1 and 2. However, we specifically wanted to test if the effect of probability format in the low-probability condition was weaker in Experiment 3 (where there was no recommendation prompt) than in Experiment 2 (where there was such a prompt).

Materials and methods

Participants

For Experiment 3, we used the same a priori power analysis recommendation as in Experiment 2. Four-hundred thirty-five English-speaking participants (220 male) holding Canadian ($n=219$) or U.S. ($n=221$) citizenship between the ages of 18 and 60 ($M=50.1$, $SD=10.0$) were recruited from Qualtrics Panels and received a credit from the Panels provider.

Design, materials, and procedure

The design, materials, and procedure of Experiment 3 were identical to Experiment 2 except that we omitted the inferred recommendation question.

Results

Confidence-weighted decision

A two-way (Probability Format $\times$ Probability Level) ANOVA on confidence-weighted decisions revealed main effects of both probability format, $F(2, 429) = 6.88$, $p = .001$, $\eta_p^2 = .031$, and level, $F(1, 429) = 61.04$, $p < .001$, $\eta_p^2 = .125$, as well as the predicted interaction effect, $F(2, 429) = 3.03$, $p = .049$, $\eta_p^2 = .014$. As the bottom row of [Figure 2](#) shows, the effect of format was only significant in the low-probability condition, $F(2, 216) = 7.73$, $p = .001$, $\eta_p^2 = .067$. Post-hoc Tukey HSD tests in the low-probability condition showed that participants were more pessimistic following verbal forecasts ($M=-4.91$, $SD=5.37$) than either numeric point ($M=-1.11$, $SD=6.53$; $p = .001$) or range ($M=-1.70$, $SD=7.03$; $p = .007$) forecasts, but that the two numeric conditions did not differ from one another ($p = .841$). Therefore, we replicated the pattern of results obtained in Experiments 1 and 2.

Effect size comparison

The successful replication allowed us to compare the effect size across Experiments 2 and 3, which had identical levels of format. Specifically, we

examined the effect of format in the low-probability condition—where format influenced our weighted decision measure—across the two experiments. To do so, we first pooled the numeric point and numeric range conditions within each experiment, as these conditions did not differ in either experiment. Next, we performed *t*-tests and calculated the effect size Cohen's *d* for each of the two experiments. The effect was significant in both Experiment 2 ($t[180.16]=4.13$, $p < .001$, $d=0.57$) and Experiment 3 ($t[182.37]=4.19$, $p < .001$, $d=0.56$). In both cases, the participants in the combined numeric conditions were more optimistic than in the verbal condition. The *z*-score of the difference in Cohen's *d* between the two experiments was 0.06, indicating that the effect size was almost identical in both experiments.

Experiment 4

Experiments 1-3 provided consistent support for the directionality account, albeit within a limited context and set of forecast terms or phrases. The purpose of Experiment 4 was twofold. First, we examined the generalizability of our findings using an organizational decision-making scenario (i.e., the CEO of a holding company), as well as a new set of verbal probability terms (i.e., *improbable* and *probable*). Second, we wanted to address the possibility that describing numeric probabilities as an 'X% *chance*' influenced their positive directional interpretation, as some research indicates the word *chance* may imply positive directionality unless negated or paired with a very low probability (Teigen & Brun, 1995, 1999). Thus, the format effect in the low-probability condition of Experiments 1-3 may be due to this potentially confounding variable. Accordingly, in Experiment 4 we also used a numeric condition in which the word *chance* did not appear.

Materials and methods

Participants

A G*Power (Faul et al., 2007) sensitivity power analysis based on the weakest interaction in Experiment 2 ($\eta_p^2 = 0.03$ for confidence-weighted decision) recommended a sample size of six-hundred and ninety-five participants. We recruited six-hundred ninety-six English-speaking participants (338 male, 352 female, 5 other, 1 prefer not to say) holding Australian ($n=11$), Canadian ($n=57$), New Zealand ($n=6$), U.K. ($n=557$), U.S. ($n=63$), dual Canadian/U.K. ($n=1$), or dual New Zealand/U.K. ($n=1$) citizenship between the ages of 18 and 60 ($M=38.6$, $SD=11.4$) were recruited using the Prolific platform and received credit for participation. To ensure high-quality data, we recruited only users with a

minimum of 100 submissions and a submission rejection rate of less than 5%.

Design

We used a 2 (Scenario: individual, organizational) $\times$ 3 (Probability Format: chance, probability, verbal) $\times$ 2 (Probability Level: low, high) between-subjects design for our experiment and subsequent analyses. The individual condition is analogous to the scenario from Experiments 1–3, whereas the organizational condition places participants in the role of a CEO of a holding company. In the verbal condition, we used the terms improbable and probable. These terms have similar directional and probabilistic properties to unlikely and likely, respectively, and are often used synonymously. For instance, the probability communication standard used by the US intelligence community treats unlikely and improbable as synonyms and likewise for the terms likely and probable (Dhimi & Mandel, 2021). The numeric chance condition is analogous to the numeric point condition in earlier experiments: *has a 30% chance* and *has a 70% chance*. The numeric probability condition conveys the probability empirically: *has a probability (in percentages) equal to 30%* and *has a probability (in percentages) equal to 70%*.

Materials and procedure

The new scenario condition and the change in verbal terminology required minor changes to the vignette, forecast, and subsequent questions. Scenario manipulations are shown in curly brackets; probability format and level manipulations are shown in square brackets:

Imagine that you {have some money/are the CEO of a holding company (i.e., a company that specializes in trading stocks)} and you are looking to invest in the stock market in the hopes of returning substantial profits. You're looking for stocks that offer favourable odds of substantial profits over the next fiscal year and you're looking to avoid stocks that don't fit this description. You consult {a financial advisor recommended to you by a close friend/the company's chief financial officer (CFO)} regarding your investment plan for the next fiscal year, and you explain your objectives. Among other information, the {advisor/CFO} draws your attention to a key forecast: **A substantial increase in the value of Bayosia Corp's stock in the next fiscal year [is improbable/is probable/has a probability (in percentages) equal to 30%/has a probability (in percentages) equal to 70%/has a 30% chance/has a 70% chance].** [emphasis in original]

The procedure, including questions, was otherwise identical to that described in Experiment 2, with the exception that the questions in the organizational condition referenced the CFO instead of the recommended advisor.

Results

Inferred recommendation

A three-way (Scenario $\times$ Probability Format $\times$ Probability Level) ANOVA on inferred recommendation revealed main effects of both probability format, $F(2, 684) = 45.42, p < .001, \eta_p^2 = .117$, and level, $F(1, 684) = 337.48, p < .001, \eta_p^2 = .330$; by contrast, the effect of scenario was not significant, $F(1, 684) = 1.36, p = .244$. Again, we find the predicted interaction effect between probability format and level, $F(2, 684) = 165.61, p < .001, \eta_p^2 = .116$; all other two- and three-way interactions were not significant, all $p \geq .567$. Figure 3 depicts the results, collapsed across the individual and organizational scenarios. We also collapsed across the scenario variable for our subsequent post hoc analyses, which began with a pair of one-way (Probability Format) univariate ANOVAs, one for each probability level condition. The analysis revealed that the effect of probability format was significant in the low-probability condition, $F(2, 342) = 51.79, p < .001, \eta_p^2 = .232$, but not the high-probability condition, $p = .742$. Post-hoc Tukey HSD within the low-probability condition ANOVA revealed that verbal

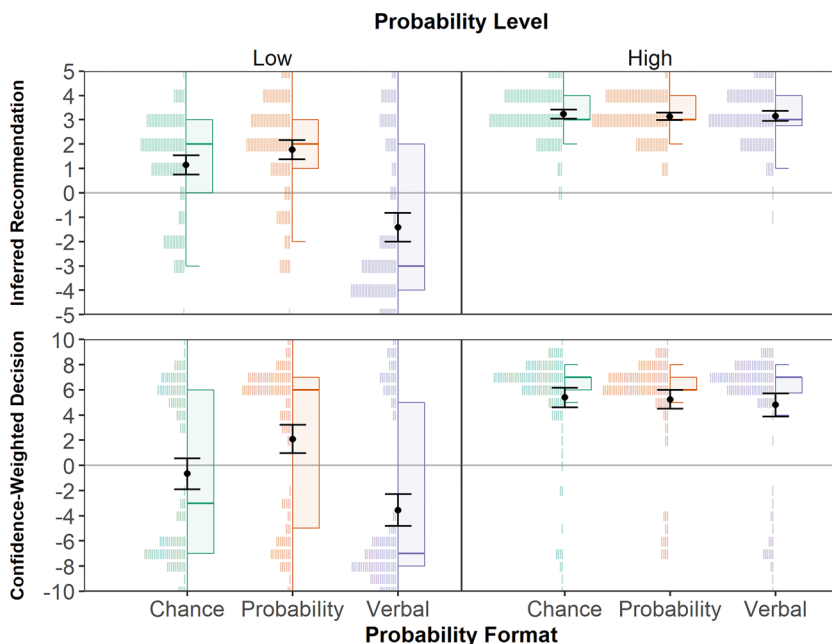


Figure 3. Effect of probability format and probability level on inferred recommendation.

Note: Histograms are sample data and each stroke represents one participant; Box-and-whiskers plots are sample data; error plots are estimated marginal means and 95% CI. The inferred recommendation scale values run from $-5 = \text{Definitely do not invest}$ to $+5 = \text{Definitely do invest}$. The confidence-weighted decision scale is a product of the decision response ($\text{no} = -1, \text{yes} = +1$) and the confidence response ($0 = \text{Not at all confident}, 10 = \text{Completely confident}$).

forecasts were inferred to be more pessimistic ($M = -1.42$, $SD = 3.17$) than either the chance format ($M = 1.15$, $SD = 2.11$, $p < .001$) or probability format ($M = 1.78$, $SD = 2.17$, $p < .001$) forecasts. The two numeric formats did not differ, $p = .145$. This pattern of results suggests that the positive directionality and subsequent optimistic interpretation of numeric forecasts in previous experiments was not simply imparted by our use of the word *chance*.

Confidence-weighted decision

We submitted the confidence-weighted decision measure to a three-way (Scenario $\times$ Probability Format $\times$ Probability Level) ANOVA, which revealed main effects of both probability format, $F(2, 684) = 17.36$, $p < .001$, $\eta_p^2 = .048$, and level, $F(1, 684) = 192.84$, $p < .001$, $\eta_p^2 = .220$; by contrast, the effect of scenario was not significant, $p = .153$. The predicted interaction between probability format and level was significant, $F(2, 684) = 12.78$, $p < .001$, $\eta_p^2 = .036$; the remaining two- and three-way interactions were not significant, all p 's $\geq .604$. Again, we collapsed across the scenario variable within [Figure 3](#) and our subsequent post-hoc analyses. As with inferred recommendation, we began with a pair of one-way (Probability Format) univariate ANOVAs, one for each probability level condition. The effect of format was significant in the low probability condition, $F(2, 342) = 21.80$, $p < .001$, $\eta_p^2 = .113$, but not the high-probability condition, $p = .575$. Post-hoc Tukey HSD within the low-probability condition ANOVA revealed that confidence-weighted decisions were significantly more pessimistic in the verbal condition ($M = -3.55$, $SD = 6.77$) than either the chance format ($M = -0.67$, $SD = 6.57$; $p = .003$) or the probability format ($M = 2.09$, $SD = 6.15$; $p < .001$). Surprisingly, participants were more pessimistic following a chance forecast than following a probability forecast, $p = .004$. Taken together, the results contradict the hypothesis that the word *chance* was responsible for the optimistic behavior in the numeric conditions of prior experiments.

General discussion

The present research clarifies how the directionality of numeric and verbal probabilities systematically differ to influence both the recommendations receivers infer from probabilistic forecasts and the ensuing decisions they make (or at least believe they would make). Across four experiments, the results consistently showed that where verbal and numeric formats were both positive (i.e., in the high-probability condition), no significant difference in inferred recommendation or decision measures was observed. Participants were optimistic about investing in both cases. Equally consistent across the experiments, we found that where directionality differs (i.e., in the low-probability condition in which only the verbal format was unambiguously negative),

participants predictably inferred more pessimistic recommendations and were more pessimistic in their choices in the verbal condition than in the numeric condition. In Experiments 1 and 2, we confirmed that participants' inferred recommendations largely mediated the effect of format on decision-making. The results of Experiment 3 reinforce our directionality account, replicating Experiment 2's confidence-weighted decision results and showing that the effect was not simply a consequence of asking participants to interpret the recommendation. Finally, the results of Experiment 4 show that the directionality account generalizes to an organizational (as opposed to individual) decision-making scenario and to a new pair of verbal probabilistic expressions. It also demonstrated that the relative positive directionality of numeric forecasts did not depend on describing the probability as a '*chance*'. Thus, our findings show that directionality, through the interaction of probability format and probability level, affects the inferences receivers make about senders' recommendations and that these inferences shape receivers' decisions.

The present findings are consistent with psycholinguistic research on the pragmatics of quantifiers (Moxey & Sanford, 1993; Sher & McKenzie, 2006) and extend work on the effects of directionality, in particular, on human reasoning and decision-making (Brun & Teigen, 1988; Teigen & Brun, 1995, 1999). Building on such work, we show that the directionality of probabilities, much like frames (Honda & Yamagishi, 2017; Jenkins & Harris, 2021; McKenzie & Nelson, 2003; Sher & McKenzie, 2006), can leak pragmatic information. In particular, directionality can serve as a construct that leads to novel, testable predictions involving the interactive effect of probability format and probability level on receivers' inferences about senders' implicit recommendations and receivers' subsequent decisions. That is, we correctly predicted that discrepancies in responses to probability information encoded as either verbal or numeric quantifiers would be pronounced at the low end of the probability continuum but not at the high end of the continuum. Thus, our account can help pinpoint where format effects are likely to be present. For instance, our account explains why Jenkins et al. (2017) did not obtain a predicted format effect on credibility assessments involving high probabilities (i.e., common directionality) even though they obtained an effect for low probabilities (i.e., divergent directionality).

Building on Teigen and Brun (2000), we have assumed that numeric probabilities *tend* to convey positive (or at least ambiguous) directionality, but we did not explicitly test that assumption in the present research. Therefore, future research could profitably examine the conditions under which numeric probabilities might take on negative directionality. Future research could also examine *incongruent* verbal terms (i.e., positive low-probability terms and negative high-probability terms). In the incongruent case, the directionality account predicts a reversal of the interaction effects observed in the present research; namely, the divergence of responses for high probabilities and the similarity of responses for low probabilities.

An interesting finding of Experiments 2 and 3 was that participants in the numeric point and numeric range conditions behaved quite similarly. An oft-touted advantage of verbal probabilities is avoiding overprecision; relatedly, an oft-touted criticism of verbal probabilities is that the probabilities are vague and ineffective at communicating uncertainty (Dhmi & Mandel, 2022; Wallsten et al., 1993). We have shown that by using numeric range forecasts experts can avoid overprecision while being clear about the level of uncertainty, obviating one of the advantages of verbal probabilities and the potential drawback of numeric point forecasts.

One potential limitation of the present research is that the alternative outcomes may not be clearly articulated or understood. Consider the case where the expert communicates the stock ‘has about a 25% chance of increasing substantially’: do participants interpret this as a 75% chance of decreasing or do they correctly account for all alternatives (e.g., decrease, stay the same, or minor increase)? We do not believe such ambiguities fundamentally challenge our conclusions. To the extent this is a limitation, it is equally present across all probability levels and formats we have used. Furthermore, the tendency of receivers to reward or punish expert judges based on the *accuracy* of the implied recommendation (Collins & Mandel, 2019) is inconsistent with the notion that potentially ambiguous interpretations of possible outcomes explain the present results. Nevertheless, research indicates that alternative outcomes are more readily imagined for numeric probabilities than for verbal probabilities; i.e., it is easier to calculate $1 - P$ given P than to imagine the alternatives to *unlikely* (Welkenhuysen et al., 2001). Thus, future studies might examine whether directionality effects are attenuated when the possible outcomes are fully explicated.

Our findings have implications for organizations that rely on or produce expert judgments. In everyday life, information leakage due to directionality may be desirable. A naïve investor planning for retirement may be better served by the implicit recommendations of a verbal forecast than by the precise probabilities of a numeric forecast. However, in areas such as national security intelligence, where analytic assessments are supposed to be policy-neutral (Kent, 1951) and transparent (Dhmi et al., 2015; Irwin & Mandel, 2019) the consequences of miscommunication can have deleterious consequences. In the worst case, both the sender and receiver are altogether unaware of the pragmatic influence of verbal probabilities, prompting irrational and inconsistent decisions. In other cases, senders could select language that intentionally nudges end-users toward particular beliefs or decisions (Mandel et al., 2021; Piercey, 2009). In yet others, translation scales such as those used by intelligence communities (for reviews see, Dhmi & Mandel, 2021, 2022; Mandel & Irwin, 2021c) may constrain the terms experts can use, forcing them to employ negative directionality terms like *unlikely* when, in fact, the communicated probability may be relatively higher than its base rate and therefore justify a more optimistic approach.

Experts and organizations should carefully consider these factors when developing their communication standards. If such communication aims to both provide probability estimates and offer guidance it might be advantageous to decouple these communicative aims using numeric probabilities to express the probability estimate and words to offer explicit recommendations or guidance. This recommendation is supported by evidence showing that people view numeric probabilities as more informative than verbal probabilities (Collins & Mandel, 2019; Irwin & Mandel, 2023). Of course, numeric estimates cannot clearly convey on their own a rationale for an estimate or a recommendation for decision-makers. Both are often needed, but at least in some important cases where unambiguous communication is vital, they should not be conflated.

Disclosure statement

No potential conflict of interest was reported by the authors.

Ethics approval

All research conducted under the approval and review of the Human Research Ethics Committee of Defence Research and Development Canada, Toronto Research Centre.

Consent to participate

Informed consent was obtained from all individual participants included in the study.

Consent for publication

Not applicable; all data used in the study are anonymized.

Availability of data and materials

The experimental materials are provided within the text and the companion Open Science Framework repository (Collins et al., 2022; <https://osf.io/evwm8/>). Data used in the study are available *via* the same.

Code availability

The code for the statistical analysis and generation of accompanying figures are provided in the companion Open Science Framework repository (Collins et al., 2022; <https://osf.io/evwm8/>).

Authors' contributions

R. N. Collins and D. R. Mandel developed the study concept. All authors contributed to the design. Survey administration and data collection were handled by Qualtrics Panels. R. N. Collins and B. A. Macleod performed the data analysis. R. N. Collins and D. R. Mandel interpreted the analysis. R. N. Collins drafted the initial manuscript, and D. R. Mandel provided critical revisions and rewrites. All authors approved the final version of the manuscript for submission.

Other acknowledgments

We thank Daniel Irwin for his assistance in the ethics approval and data collection processes.

Open practices statement

All material and data for the experiments reported here are available *via* an Open Science Framework repository (Collins et al., 2022; <https://osf.io/evwm8/>). The experiments were not pre-registered.

Funding

This work was supported by Canadian Safety and Security Program project CSSP-2018-TI-2394.

ORCID

Robert N. Collins  <http://orcid.org/0000-0002-1714-7215>

David R. Mandel  <http://orcid.org/0000-0003-1036-2286>

References

- Barnes, A. (2016). Making intelligence analysis more intelligent: Using numeric probabilities. *Intelligence and National Security*, 31(3), 327–344. <https://doi.org/10.1080/02684527.2014.994955>
- Beyth-Marom, R. (1982). How probable is probable? A numerical translation of verbal probability expressions. *Journal of Forecasting*, 1(3), 257–269. <https://doi.org/10.1002/for.3980010305>
- Brun, W., & Teigen, K. H. (1988). Verbal probabilities: Ambiguous, context-dependent, or both? *Organizational Behavior and Human Decision Processes*, 41(3), 390–404. [https://doi.org/10.1016/0749-5978\(88\)90036-2](https://doi.org/10.1016/0749-5978(88)90036-2)
- Budescu, D. V., Broomell, S., & Por, H.-H. (2009). Improving communication of uncertainty in the reports of the intergovernmental panel on climate change. *Psychological Science*, 20(3), 299–308. <https://doi.org/10.1111/j.1467-9280.2009.02284.x>
- Budescu, D. V., Karelitz, T. M., & Wallsten, T. S. (2003). Predicting the directionality of probability words from their membership functions. *Journal of Behavioral Decision Making*, 16(3), 159–180. <https://doi.org/10.1002/bdm.440>

- Budescu, D. V., Por, H. H., Broomell, S. B., & Smithson, M. (2014). The interpretation of IPCC probabilistic statements around the world. *Nature Climate Change*, 4(6), 508–512. <https://doi.org/10.1038/nclimate2194>
- Collins, R. N., & Mandel, D. R. (2019). Cultivating credibility with probability words and numbers. *Judgment and Decision Making*, 14(6), 683–695. <http://journal.sjdm.org/19/190912/jdm190912.html> <https://doi.org/10.1017/S1930297500005404>
- Collins, R. N., Mandel, D. R., & Macleod, B. (2022). Verbal and numeric probabilities differentially shape decisions. [Open Science Framework project] <https://doi.org/10.17605/OSF.IO/EVWM8>
- Dhami, M. K., & Mandel, D. R. (2021). Words or numbers? Communicating probability in intelligence analysis. *The American Psychologist*, 76(3), 549–560. <https://doi.org/10.1037/amp0000637>
- Dhami, M. K., & Mandel, D. R. (2022). Communicating uncertainty using words and numbers. *Trends in Cognitive Sciences*, 26(6), 514–526. <https://doi.org/10.1016/j.tics.2022.03.002>
- Dhami, M. K., Mandel, D. R., Mellers, B. A., & Tetlock, P. E. (2015). Improving intelligence analysis with decision science. *Perspectives on Psychological Science: A Journal of the Association for Psychological Science*, 10(6), 753–757. <https://doi.org/10.1177/1745691615598511>
- Dieckmann, N. F., Mauro, R., & Slovic, P. (2010). The effects of presenting imprecise probabilities in intelligence forecasts. *Risk Analysis: An Official Publication of the Society for Risk Analysis*, 30(6), 987–1001. <https://doi.org/10.1111/j.1539-6924.2010.01384.x>
- Erev, I., & Cohen, B. L. (1990). Verbal versus numerical probabilities: Efficiency, biases, and the preference paradox. *Organizational Behavior and Human Decision Processes*, 45(1), 1–18. [https://doi.org/10.1016/0749-5978\(90\)90002-Q](https://doi.org/10.1016/0749-5978(90)90002-Q)
- Faul, F., Erdfelder, E., Lang, A. G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39, 175–191. <https://doi.org/10.3758/BF03193146>
- Fisher, S., & Mandel, D. R. (2021). Risky-choice framing and rational decision-making. *Philosophy Compass*, 16(8), e12763. <https://doi.org/10.1111/phc3.12763>
- Funtowicz, S. O., & Ravetz, J. R. (1990). *Uncertainty and quality in science for policy*. Springer. <https://doi.org/10.1007/978-94-009-0621-1>
- Hayes, A. F. (2022). *Introduction to mediation, moderation, and conditional process analysis, third edition: A regression-based approach*. Guilford Publications.
- Hilton, D. J. (2011). Linguistic polarity, outcome framing, and the structure of decision making: A pragmatic approach. In G. Keren (Ed.), *Perspectives on framing* (pp. 135–155). Psychology Press.
- Ho, E. H., Budescu, D. V., Dhami, M. K., & Mandel, D. R. (2015). Improving the communication of uncertainty in climate science and intelligence analysis. *Behavioral Science & Policy*, 1(2), 43–55. <https://doi.org/10.1353/bsp.2015.0015>
- Honda, H., & Yamagishi, K. (2006). Directional verbal probabilities: Inconsistencies between preferential judgments and numerical meanings. *Experimental Psychology*, 53(3), 161–170. <https://doi.org/10.1027/1618-3169.53.3.161>
- Honda, H., & Yamagishi, K. (2009). Perceived certainty based on verbal probability phrases: Effect of directionality and its dependence on method. *Japanese Psychological Research*, 51(4), 266–273. <https://doi.org/10.1111/j.1468-5884.2009.00409.x>
- Honda, H., & Yamagishi, K. (2017). Communicative functions of directional verbal probabilities: Speaker's choice, listener's inference, and reference points. *Quarterly*

- Journal of Experimental Psychology*, 70(10), 2141–2158. <https://doi.org/10.1080/17470218.2016.1225779>
- Irwin, D., & Mandel, D. R. (2019). Improving information evaluation for intelligence production. *Intelligence and National Security*, 34(4), 503–525. <https://doi.org/10.1080/02684527.2019.1569343>
- Irwin, D., & Mandel, D. R. (2023). Communicating uncertainty in national security intelligence: Expert and non-expert interpretations of and preferences for verbal and numeric formats. *Risk Analysis*, 43(5), 943–957. <https://doi.org/10.1111/risa.14009>
- Jenkins, S. C., & Harris, A. J. L. (2021). Maintaining credibility when communicating uncertainty: The role of directionality. *Thinking & Reasoning*, 27(1), 97–123. <https://doi.org/10.1080/13546783.2020.1723694>
- Jenkins, S. C., Harris, A. J. L., & Lark, R. M. (2017). Maintaining credibility when communicating uncertainty: The role of communication format. In G. Gunzelmann, A. Howes, T. Tenbrink, & E. J. Davelaar (Eds.), *Proceedings of the 39th Annual Conference of the Cognitive Science Society* (pp. 582–587). Cognitive Science Society.
- Juanchich, M., & Sirota, M. (2020). Do people really prefer verbal probabilities? *Psychological Research*, 84(8), 2325–2338. <https://doi.org/10.1007/s00426-019-01207-0>
- Juanchich, M., Sirota, M., & Butler, C. L. (2012). The perceived functions of linguistic risk quantifiers and their effect on risk, negativity perception and decision making. *Organizational Behavior and Human Decision Processes*, 118(1), 72–81. <https://doi.org/10.1016/j.obhdp.2012.01.002>
- Juanchich, M., Sirota, M., Karelitz, T. M., & Villejoubert, G. (2013). Can membership-functions capture the directionality of verbal probabilities? *Thinking & Reasoning*, 19(2), 231–247. <https://doi.org/10.1080/13546783.2013.772538>
- Kent, S. (1951). *Strategic intelligence for American world policy*. Princeton University Press.
- Ligertwood, A., & Edmond, G. (2012). Expressing evaluative forensic science opinions in a court of law. *Law, Probability and Risk*, 11(4), 289–302. <https://doi.org/10.1093/lpr/mgs016>
- Lipkus, I. M. (2007). Numeric, verbal, and visual formats of conveying health risks: Suggested best practices and future recommendations. *Medical Decision Making: An International Journal of the Society for Medical Decision Making*, 27(5), 696–713. <https://doi.org/10.1177/0272989X07307271>
- Mandel, D. R. (2014). Do framing effects reveal irrational choice? *Journal of Experimental Psychology. General*, 143(3), 1185–1198. <https://doi.org/10.1037/a0034207>
- Mandel, D. R. (2022). Framing, equivalence, and rational inference. *The Behavioral and Brain Sciences*, 45, e234. <https://doi.org/10.1017/S0140525X22000954>
- Mandel, D. R., & Irwin, D. (2021a). Facilitating sender-receiver agreement in communicated probabilities: Is it best to use words, numbers or both? *Judgment and Decision Making*, 16(2), 363–393. <http://journal.sjdm.org/20/200930/jdm200930.html> <https://doi.org/10.1017/S1930297500008603>
- Mandel, D. R., & Irwin, D. (2021b). On measuring agreement with numerically bounded linguistic probability schemes: A re-analysis of data from Wintle, Fraser, Wills, Nicholson, and Fidler (2019). *PloS One*, 16(3), e0248424. <https://doi.org/10.1371/journal.pone.0248424>

- Mandel, D. R., & Irwin, D. (2021c). Uncertainty, intelligence, and national security decisionmaking. *International Journal of Intelligence and CounterIntelligence*, 34(3), 558–582. <https://doi.org/10.1080/08850607.2020.1809056>
- Mandel, D. R., Wallsten, T. S., & Budescu, D. V. (2021). Numerically bounded linguistic probability schemes are unlikely to communicate uncertainty effectively. *Earth's Future*, 9(1), 5. <https://doi.org/10.1029/2020EF001526>
- Mazur, D. J., & Merz, J. F. (1994). Patients' interpretations of verbal expressions of probability: Implications for securing informed consent to medical interventions. *Behavioral Sciences & the Law*, 12(4), 417–426. <https://doi.org/10.1002/bsl.2370120410>
- McKenzie, C. R., & Nelson, J. D. (2003). What a speaker's choice of frame reveals: Reference points, frame selection, and framing effects. *Psychonomic Bulletin & Review*, 10(3), 596–602. <https://doi.org/10.3758/bf03196520>
- Mellers, B. A., Baker, J. D., Chen, E., Mandel, D. R., & Tetlock, P. E. (2017). How generalizable is good judgment? A multi-task, multi-benchmark study. *Judgment and Decision Making*, 12(4), 369–381. <http://journal.sjdm.org/17/17408/jdm17408.pdf> <https://doi.org/10.1017/S1930297500006240>
- Mosteller, F., & Youtz, C. (1990). Quantifying probabilistic expressions. *Statistical Science*, 5(1), 2–12. <http://www.jstor.org/stable/2245869> <https://doi.org/10.1214/ss/1177012242>
- Moxey, L. M., & Sanford, A. J. (1993). *Communicating quantities: A psychological perspective* (pp. xii, 130). Lawrence Erlbaum Associates, Inc.
- Peters, E., & Levin, I. P. (2008). Dissecting the risky-choice framing effect: Numeracy as an individual-difference factor in weighting risky and riskless options. *Judgment and Decision Making*, 3(6), 435–448. <https://doi.org/10.1017/S1930297500000012>
- Piercey, M. D. (2009). Motivated reasoning and verbal vs. numeric probability assessment: Evidence from an accounting context. *Organizational Behavior and Human Decision Processes*, 108(2), 330–341. <https://doi.org/10.1016/j.obhdp.2008.05.004>
- Schmeltzer, C., & Hilton, D. J. (2014). To do or not to do? A cognitive consistency model for drawing conclusions from conditional instructions and advice. *Thinking & Reasoning*, 20(1), 16–50. <https://doi.org/10.1080/13546783.2013.798594>
- Sher, S., & McKenzie, C. R. M. (2006). Information leakage from logically equivalent frames. *Cognition*, 101(3), 467–494. <https://doi.org/10.1016/j.cognition.2005.11.001>
- Teigen, K. H., & Brun, W. (1995). Yes, but it is uncertain: Direction and communicative intention of verbal probabilistic terms. *Acta Psychologica*, 88(3), 233–258. [https://doi.org/10.1016/0001-6918\(93\)E0071-9](https://doi.org/10.1016/0001-6918(93)E0071-9)
- Teigen, K. H., & Brun, W. (1999). The directionality of verbal probability expressions: Effects on decisions, predictions, and probabilistic reasoning. *Organizational Behavior and Human Decision Processes*, 80(2), 155–190. <https://doi.org/10.1006/obhd.1999.2857>
- Teigen, K. H., & Brun, W. (2000). Ambiguous probabilities: When does $p = 0.3$ reflect a possibility, and when does it express a doubt? *Journal of Behavioral Decision Making*, 13(3), 345–362. [https://doi.org/10.1002/1099-0771\(200007/09\)13:3<345::AID-BDM358>3.0.CO;2-U](https://doi.org/10.1002/1099-0771(200007/09)13:3<345::AID-BDM358>3.0.CO;2-U)
- Theil, M. (2002). The role of translations of verbal into numeric probability expressions in risk management: a meta-analysis. *Journal of Risk Research*, 5(2), 177–186. <https://doi.org/10.1080/13669870110038179>
- Tversky, A., & Kahneman, D. (1986). The framing of decisions and the evaluation of prospects. In R. Barcan Marcus, G. J. W. Dorn, & P. Weingartner (Eds.), *Studies*

- in logic and the foundations of mathematics* (Vol. 114, pp. 503–520). Elsevier. [https://doi.org/10.1016/S0049-237X\(09\)70710-4](https://doi.org/10.1016/S0049-237X(09)70710-4)
- Wallsten, T. S., & Budescu, D. V. (1995). A review of human linguistic probability processing: General principles and empirical evidence. *The Knowledge Engineering Review*, 10(1), 43–62. <https://doi.org/10.1017/S0269888900007256>
- Wallsten, T. S., Budescu, D. V., Zwick, R., & Kemp, S. M. (1993). Preferences and reasons for communicating probabilistic information in verbal or numerical terms. *Bulletin of the Psychonomic Society*, 31(2), 135–138. <https://doi.org/10.3758/BF03334162>
- Welkenhuysen, M., Evers-Kiebooms, G., & d'Ydewalle, G. (2001). The language of uncertainty in genetic risk communication: framing and verbal versus numerical information. *Patient Education and Counseling*, 43(2), 179–187. [https://doi.org/10.1016/S0738-3991\(00\)00161-0](https://doi.org/10.1016/S0738-3991(00)00161-0)
- Wintle, B. C., Fraser, H., Wills, B. C., Nicholson, A. E., & Fidler, F. (2019). Verbal probabilities: Very likely to be somewhat more confusing than numbers. *Plos One*, 14(4), e0213522. <https://doi.org/10.1371/journal.pone.0213522>